

ECE 332: Engineering Electromagnetics II

Lecture 12

Mark A. Martin

Portland State University

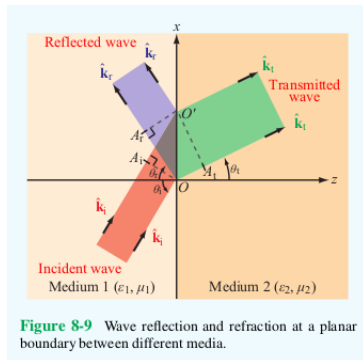
May 12, 2026

Quick Review

Snell's Laws

For **lossless** media, when the incident wave is traveling in a direction that isn't normal to the boundary, the rays of the waves **travel in the directions below** and **intersect the boundary at the given angles relative to the normal to the boundary**.

- **incident wave:**
direction $\hat{\mathbf{k}}_i$, angle $\theta_i \in [0, \pi/2]$,
- **reflected wave:**
direction $\hat{\mathbf{k}}_r$, angle $\theta_r \in [0, \pi/2]$,
- **transmitted wave:**
direction $\hat{\mathbf{k}}_t$, angle $\theta_t \in [0, \pi/2]$.



Quick Review (Continued)

- The **index of refraction** of a medium is defined as

$$n = \frac{c}{u_p} = \sqrt{\mu_r \epsilon_r} > 0. \quad (1)$$

- Under these circumstances,

$$\theta_r = \theta_i \quad (\text{Snell's law of } \mathbf{reflection}) \quad (2)$$

$$\frac{\sin \theta_t}{\sin \theta_i} = \frac{n_1}{n_2} = \frac{u_{p2}}{u_{p1}} = \sqrt{\frac{\mu_1 \epsilon_1}{\mu_2 \epsilon_2}} = \sqrt{\frac{\mu_{r1} \epsilon_{r1}}{\mu_{r2} \epsilon_{r2}}} \quad (\text{Snell's law of } \mathbf{refraction}). \quad (3)$$

- For **nonmagnetic materials**, the **law of refraction** is

$$\frac{\sin \theta_t}{\sin \theta_i} = \sqrt{\frac{\epsilon_{r1}}{\epsilon_{r2}}} = \frac{\eta_2}{\eta_1}. \quad (4)$$

- A nonmagnetic material with a **higher index of refraction** is often called **denser** than a nonmagnetic material with a **lower index**.

Quick Review (Continued)

- When $\theta_i \neq 0$ and $n_2 > n_1$, $\theta_t < \theta_i$, which is called **inward refraction**, and, when $\theta_i \neq 0$ and $n_1 > n_2$, $\theta_t > \theta_i$, which is called **outward refraction**.
- When $n_1 > n_2$, $\theta_t = \pi/2$ when θ_i is at the **critical angle** θ_c , given by

$$\sin \theta_c = n_2/n_1. \quad (5)$$

- If $\mu_1 = \mu_2$,

$$\sin \theta_c = \sqrt{\epsilon_{r2}/\epsilon_{r1}}. \quad (6)$$

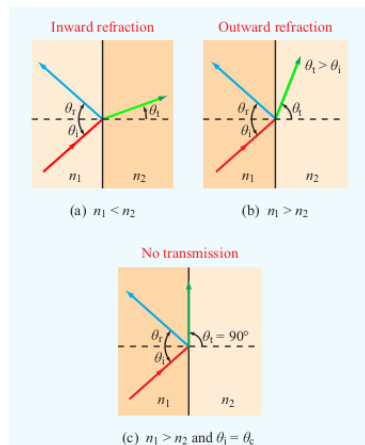
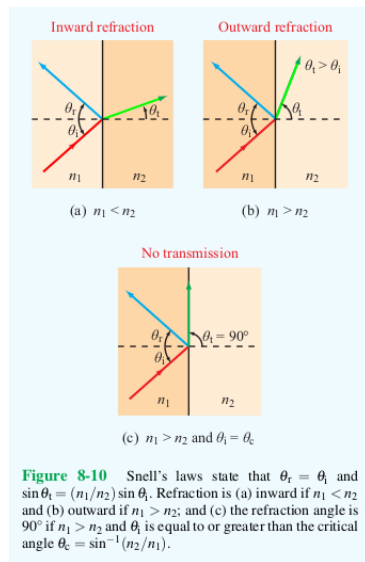


Figure 8-10 Snell's laws state that $\theta_r = \theta_i$ and $\sin \theta_t = (n_1/n_2) \sin \theta_i$. Refraction is (a) inward if $n_1 < n_2$ and (b) outward if $n_1 > n_2$; and (c) the refraction angle is 90° if $n_1 > n_2$ and θ_i is equal to or greater than the critical angle $\theta_c = \sin^{-1}(n_2/n_1)$.

Quick Review (Continued)

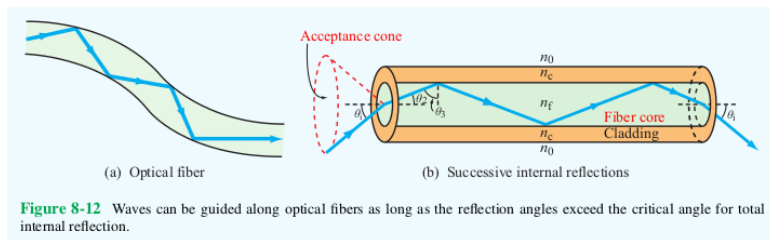
- When $n_1 > n_2$ and $\theta_i \geq \theta_c$, the incident wave is entirely reflected, which is called **total internal reflection**.



Quick Review (Continued)

Fiber Optics

Total internal reflection can be used to create small-diameter cylinders of dielectric material, typically glass or plastic, that efficiently carry light.



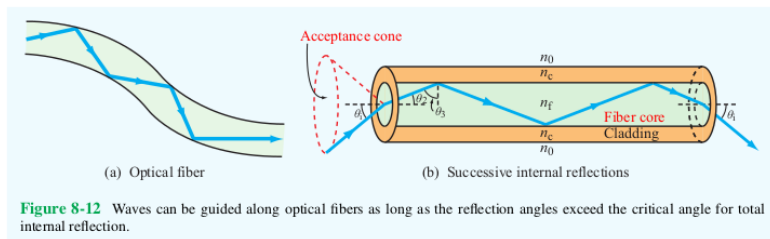
- A fiber consists of an inner **fiber core** with an index of refraction n_f that is higher than the index of refraction n_c of the surrounding **cladding**.

Quick Review (Continued)

- Light entering the end of the fiber core must have an angle of incidence θ_i satisfying

$$\sin \theta_i \leq \frac{1}{n_0} \left(n_f^2 - n_c^2 \right)^{1/2}, \quad (7)$$

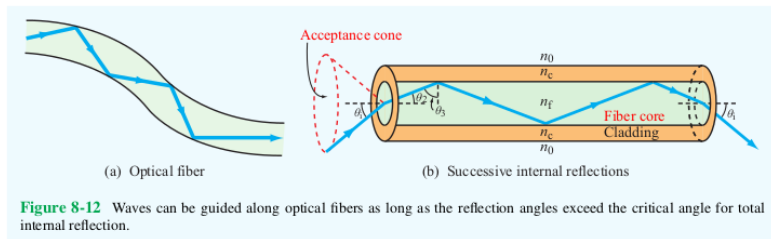
where n_0 is the index of refraction of the medium outside of the end of the fiber, for the ray to be totally internally reflected.



Quick Review (Continued)

- The **largest angle** θ_i that produces total internal reflection in the fiber is the **acceptance angle** θ_a , given by

$$\sin \theta_a = \frac{1}{n_0} \left(n_f^2 - n_c^2 \right)^{1/2}. \quad (8)$$



- The cone containing all rays that intersect the fiber core with $\theta_i \leq \theta_a$ is called the **acceptance cone**.

Quick Review (Continued)

- Rays entering the fiber core at different angles θ_i follow different paths through the fiber. Each path is called a **mode**.
- Different modes travel through the fiber in different amounts of time. This is called **modal dispersion** and distorts pulses sent through the fiber.

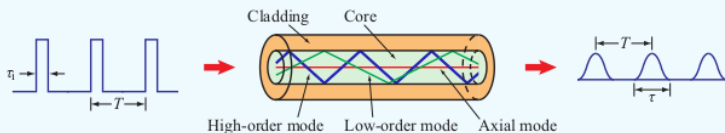


Figure 8-13 Distortion of rectangular pulses caused by modal dispersion in optical fibers.

- Rectangular pulses of initial width τ_i and period T are spread out into rounded pulses of width τ and pulses are usually created with a period $T \geq 2\tau$ to make sure it's possible to distinguish them.

Quick Review (Continued)

- For a fiber of length ℓ , the **maximum time delay** will be

$$\tau = \frac{\ell n_f}{c} \left(\frac{n_f - 1}{n_c} \right) \quad (9)$$

and the **maximum data transmission rate** meeting the requirement $T \geq 2\tau$ will then be

$$f_p = \frac{1}{T} = \frac{1}{2\tau} = \frac{cn_c}{2\ell n_f(n_f - n_c)}. \quad (10)$$

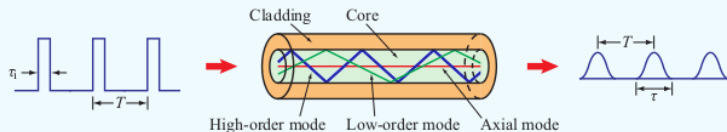
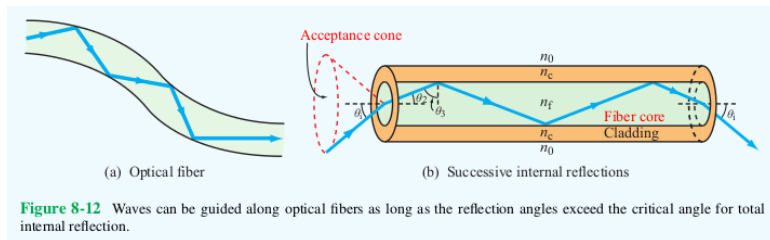


Figure 8-13 Distortion of rectangular pulses caused by modal dispersion in optical fibers.

Quick Review (Continued)

- The cone consisting of all of the paths that light at the end of the fiber is allowed to follow is called the **entrance cone**.



Quick Review (Continued)

- If the entrance cone can be restricted to be narrower than the acceptance cone, with maximum angle $\theta_i(\text{max})$, then the **maximum data transmission rate** can be **increased** to

$$f_p = \frac{c}{2\ell n_f} \left(\frac{\cos \theta_2(\text{max})}{1 - \cos \theta_2(\text{max})} \right), \quad (11)$$

where

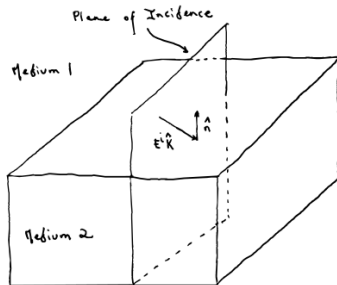
$$\cos \theta_2(\text{max}) = \left(1 - \left(\frac{n_0}{n_f} \right)^2 \sin^2 \theta_i(\text{max}) \right)^{1/2}. \quad (12)$$

Quick Review (Continued)

Reflection and Transmission at Oblique Incidence

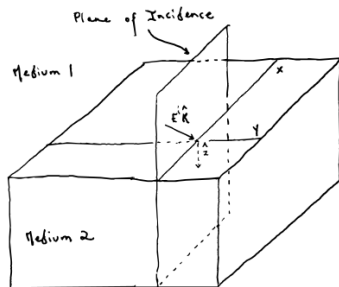
A **more complete picture** of what happens when an **EM wave encounters a boundary between two lossless media** is based on representing the wave as a **sum of two components** – a component ($\mathbf{E}_\perp, \mathbf{H}_\perp$) polarized **perpendicular to the plane of incidence** and a component ($\mathbf{E}_\parallel, \mathbf{H}_\parallel$) polarized **parallel to the plane of incidence**.

The **plane of incidence** is the plane formed by the **normal to the boundary** $\hat{\mathbf{n}}$, the **direction of propagation** $\hat{\mathbf{k}}$ of the EM wave, and the **point where a ray of the wave intersects the boundary**.



Quick Review (Continued)

- The plane of incidence, the boundary between the media, and a point where a ray of the wave intersects the boundary can be used to define a **global Cartesian coordinate system**.



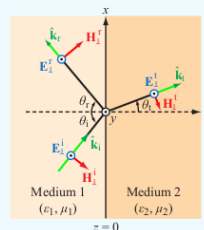
Quick Review (Continued)

- The wave $(\mathbf{E}_{\parallel}, \mathbf{H}_{\parallel})$ is called **transverse magnetic (TM)** since $\mathbf{H}_{\parallel}$ is $\perp$ the plane of incidence and $(\mathbf{E}_{\perp}, \mathbf{H}_{\perp})$ is called **transverse electric (TE)** since $\mathbf{E}_{\perp}$ is $\perp$ the plane of incidence.
- Define $(\mathbf{E}^i, \mathbf{H}^i)$ to be the EM fields of the **incident wave**, $(\mathbf{E}^r, \mathbf{H}^r)$ to be the EM fields of the **reflected wave**, and $(\mathbf{E}^t, \mathbf{H}^t)$ to be the EM fields of the **transmitted wave**. Then

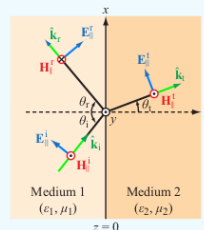
$$\mathbf{E}^i = \mathbf{E}_{\parallel}^i + \mathbf{E}_{\perp}^i, \quad \mathbf{H}^i = \mathbf{H}_{\parallel}^i + \mathbf{H}_{\perp}^i, \quad (13)$$

$$\mathbf{E}^r = \mathbf{E}_{\parallel}^r + \mathbf{E}_{\perp}^r, \quad \mathbf{H}^r = \mathbf{H}_{\parallel}^r + \mathbf{H}_{\perp}^r, \quad (14)$$

$$\mathbf{E}^t = \mathbf{E}_{\parallel}^t + \mathbf{E}_{\perp}^t, \quad \mathbf{H}^t = \mathbf{H}_{\parallel}^t + \mathbf{H}_{\perp}^t. \quad (15)$$



(a) Perpendicular polarization

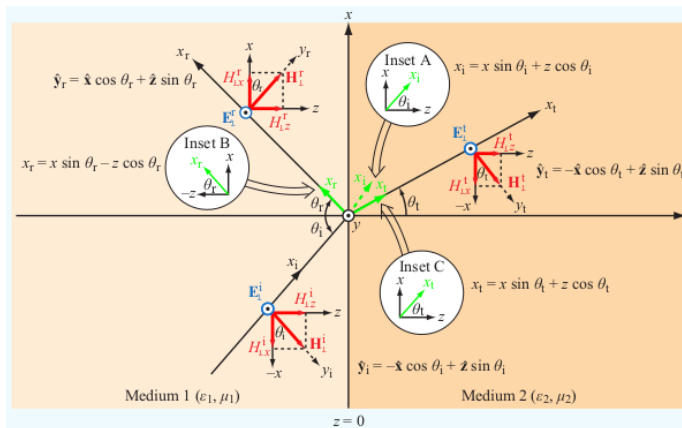


(b) Parallel polarization

Quick Review (Continued)

Perpendicular Polarization

We first considered an incident wave ($\mathbf{E}_\perp^i, \mathbf{H}_\perp^i$) that is **polarized perpendicular** to the plane of incidence.



Quick Review (Continued)

- The relationships between the amplitudes of the reflected and transmitted waves, $E_{\perp 0}^r$ and $E_{\perp 0}^t$, and the angles of reflection and transmission, θ_r , and θ_t , and the 2 parameters $E_{\perp 0}^i$ and θ_i were determined using the **tangential boundary conditions**.
- These imply the **phase-matching condition**

$$k_1 \sin \theta_i = k_1 \sin \theta_r = k_2 \sin \theta_t. \quad (16)$$

- The phase-matching condition implies **Snell's law of reflection**

$$\theta_r = \theta_i \quad (17)$$

and **Snell's law of refraction**

$$\frac{\sin \theta_t}{\sin \theta_i} = \frac{k_1}{k_2} = \sqrt{\frac{\mu_1 \epsilon_1}{\mu_2 \epsilon_2}} = \frac{n_1}{n_2}. \quad (18)$$

Quick Review (Continued)

Altogether, when the incident wave ($\mathbf{E}_{\perp}^i, \mathbf{H}_{\perp}^i$) is **polarized perpendicular** to the plane of incidence,

- the phasors of the EM fields of the **incident wave** are

$$\tilde{\mathbf{E}}_{\perp}^i = E_{\perp 0}^i e^{-jk_1(x \sin \theta_i + z \cos \theta_i)} \hat{\mathbf{y}}, \quad (19)$$

$$\tilde{\mathbf{H}}_{\perp}^i = \frac{E_{\perp 0}^i}{\eta_1} e^{-jk_1(x \sin \theta_i + z \cos \theta_i)} (-\hat{\mathbf{x}} \cos \theta_i + \hat{\mathbf{z}} \sin \theta_i), \quad (20)$$

- the phasors of the EM fields of the **reflected wave** are

$$\tilde{\mathbf{E}}_{\perp}^r = \Gamma_{\perp} E_{\perp 0}^i e^{-jk_1(x \sin \theta_i - z \cos \theta_i)} \hat{\mathbf{y}}, \quad (21)$$

$$\tilde{\mathbf{H}}_{\perp}^r = \Gamma_{\perp} \frac{E_{\perp 0}^i}{\eta_1} e^{-jk_1(x \sin \theta_i - z \cos \theta_i)} (\hat{\mathbf{x}} \cos \theta_i + \hat{\mathbf{z}} \sin \theta_i), \quad (22)$$

where

$$\Gamma_{\perp} = \frac{\eta_2 \cos \theta_i - \eta_1 \cos \theta_t}{\eta_2 \cos \theta_i + \eta_1 \cos \theta_t}, \quad (23)$$

is the **Fresnel reflection coefficient for perpendicular polarization**.

Quick Review (Continued)

- and the phasors of the the EM fields of the **transmitted wave** are

$$\tilde{\mathbf{E}}_{\perp}^t = \tau_{\perp} E_{\perp 0}^i e^{-jk_2(x \sin \theta_t + z \cos \theta_t)} \hat{\mathbf{y}}, \quad (24)$$

$$\tilde{\mathbf{H}}_{\perp}^t = \tau_{\perp} \frac{E_{\perp 0}^i}{\eta_2} e^{-jk_2(x \sin \theta_t + z \cos \theta_t)} (-\hat{\mathbf{x}} \cos \theta_t + \hat{\mathbf{z}} \sin \theta_t), \quad (25)$$

where

$$\tau_{\perp} = \frac{2\eta_2 \cos \theta_i}{\eta_2 \cos \theta_i + \eta_1 \cos \theta_t}, \quad (26)$$

is the **Fresnel transmission coefficient for perpendicular polarization**.

- Reflecting the **balance of power across the boundary**

$$\tau_{\perp} = 1 + \Gamma_{\perp}. \quad (27)$$

Quick Review (Continued)

Special cases:

- When **Medium 2 is a perfect conductor**, $\eta_2 = 0$, $\Gamma_{\perp} = -1$, and $\tau_{\perp} = 0$. So, the wave is perfectly reflected.
- When the media are **nonmagnetic**, $\mu_1 = \mu_2 = \mu_0$ and

$$\Gamma_{\perp} = \frac{\cos \theta_i - \sqrt{(\epsilon_2/\epsilon_1) - \sin^2 \theta_i}}{\cos \theta_i + \sqrt{(\epsilon_2/\epsilon_1) - \sin^2 \theta_i}}. \quad (28)$$

Here, $\epsilon_2/\epsilon_1 = (n_2/n_1)^2$. So, $\Gamma_{\perp}$ can be written in terms of the indices of refraction.

Reflection and Transmission at Oblique Incidence (Cont.)

Parallel Polarization

For an incident wave ($\mathbf{E}_{\parallel}^i, \mathbf{H}_{\parallel}^i$) that is **polarized parallel** to the plane of incidence, the EM fields of the **incident wave** are

$$\tilde{\mathbf{E}}_{\parallel}^i = E_{\parallel 0}^i e^{-jk_1(x \sin \theta_i + z \cos \theta_i)} \cdot (\hat{\mathbf{x}} \cos \theta_i - \hat{\mathbf{z}} \sin \theta_i), \quad (29)$$

$$\tilde{\mathbf{H}}_{\parallel}^i = \frac{E_{\parallel 0}^i}{\eta_1} e^{-jk_1(x \sin \theta_i + z \cos \theta_i)} \hat{\mathbf{y}}. \quad (30)$$

The **tangential boundary conditions** imply Snell's laws and relationships between the amplitudes and angles of the reflected and transmitted waves and those of the incident wave.

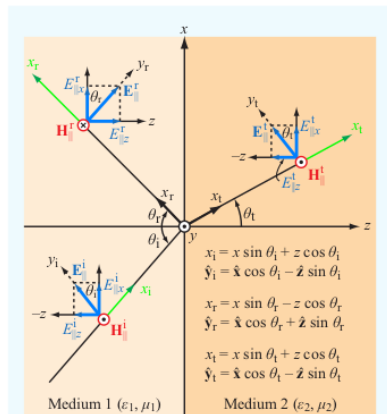


Figure 8-16 Parallel-polarized plane wave incident at an angle θ_i upon a planar boundary.

Reflection and Transmission at Oblique Incidence (Cont.)

The phasors of the EM fields of the **reflected wave** are

$$\tilde{\mathbf{E}}_{\parallel}^r = \Gamma_{\parallel} E_{\parallel 0}^i e^{-jk_1(x \sin \theta_i - z \cos \theta_i)} (\hat{\mathbf{x}} \cos \theta_i + \hat{\mathbf{z}} \sin \theta_i), \quad (31)$$

$$\tilde{\mathbf{H}}_{\parallel}^r = -\Gamma_{\parallel} \frac{E_{\parallel 0}^i}{\eta_1} e^{-jk_1(x \sin \theta_i - z \cos \theta_i)} \hat{\mathbf{y}}, \quad (32)$$

and the phasors of the EM fields of the **transmitted wave** are

$$\tilde{\mathbf{E}}_{\parallel}^t = \tau_{\parallel} E_{\parallel 0}^i e^{-jk_2(x \sin \theta_t + z \cos \theta_t)} (\hat{\mathbf{x}} \cos \theta_t - \hat{\mathbf{z}} \sin \theta_t), \quad (33)$$

$$\tilde{\mathbf{H}}_{\parallel}^t = \tau_{\parallel} \frac{E_{\parallel 0}^i}{\eta_2} e^{-jk_2(x \sin \theta_t + z \cos \theta_t)} \hat{\mathbf{y}}, \quad (34)$$

where the **Fresnel reflection coefficient for parallel polarization** is

$$\Gamma_{\parallel} = \frac{\eta_2 \cos \theta_t - \eta_1 \cos \theta_i}{\eta_2 \cos \theta_t + \eta_1 \cos \theta_i}, \quad (35)$$

Reflection and Transmission at Oblique Incidence (Cont.)

the **Fresnel transmission coefficient for parallel polarization** is

$$\tau_{\parallel} = \frac{2\eta_2 \cos \theta_i}{\eta_2 \cos \theta_t + \eta_1 \cos \theta_i}, \quad (36)$$

and **Snell's law of refraction** implies

$$\frac{\sin \theta_t}{\sin \theta_i} = \frac{k_1}{k_2} = \sqrt{\frac{\mu_1 \epsilon_1}{\mu_2 \epsilon_2}} = \frac{n_1}{n_2}. \quad (37)$$

Note that the boundary conditions also imply **Snell's law of reflection**

$$\theta_r = \theta_i. \quad (38)$$

In this case,

$$\tau_{\parallel} = (1 + \Gamma_{\parallel}) \frac{\cos \theta_i}{\cos \theta_t}. \quad (39)$$

Reflection and Transmission at Oblique Incidence (Cont.)

Special cases:

- When **Medium 2 is a perfect conductor**, $\eta_2 = 0$ and $\Gamma_{\parallel} = -1$ and $\tau_{\parallel} = 0$.
- When the media are **nonmagnetic**, $\mu_1 = \mu_2 = \mu_0$ and

$$\Gamma_{\parallel} = \frac{-(\epsilon_2/\epsilon_1) \cos \theta_i - \sqrt{(\epsilon_2/\epsilon_1) - \sin^2 \theta_i}}{(\epsilon_2/\epsilon_1) \cos \theta_i + \sqrt{(\epsilon_2/\epsilon_1) - \sin^2 \theta_i}}. \quad (40)$$

Questions:

Do the **angles** θ_r and θ_t **differ** for $\mathbf{E}_{\perp}$ and $\mathbf{E}_{\parallel}$? **No**, they're the same.
Why? Snell's laws apply in both cases.

Reflection and Transmission at Oblique Incidence (Cont.)

Recall that

$$\mathbf{E}^i = \mathbf{E}_{\parallel}^i + \mathbf{E}_{\perp}^i, \quad \mathbf{H}^i = \mathbf{H}_{\parallel}^i + \mathbf{H}_{\perp}^i, \quad (41)$$

$$\mathbf{E}^r = \mathbf{E}_{\parallel}^r + \mathbf{E}_{\perp}^r, \quad \mathbf{H}^r = \mathbf{H}_{\parallel}^r + \mathbf{H}_{\perp}^r, \quad (42)$$

$$\mathbf{E}^t = \mathbf{E}_{\parallel}^t + \mathbf{E}_{\perp}^t, \quad \mathbf{H}^t = \mathbf{H}_{\parallel}^t + \mathbf{H}_{\perp}^t. \quad (43)$$

Engineering Questions:

- What would we **need to know** to determine $\mathbf{E}^r$ and $\mathbf{E}^t$?

We would need to know the **incident amplitudes** $E_{\perp 0}^i$ and $E_{\parallel 0}^i$, **incident angle** θ_i , the **angular frequency** ω of the EM wave, and the **constitutive parameters** μ_1 , μ_2 , ϵ_1 , and ϵ_2 .

- Which of these parameters would be **harder to determine** and which would be easier?

$E_{\perp 0}^i$ and $E_{\parallel 0}^i$ would probably be harder and θ_i , ω , μ_1 , μ_2 , ϵ_1 , and ϵ_2 would probably be easier.

Reflection and Transmission at Oblique Incidence (Cont.)

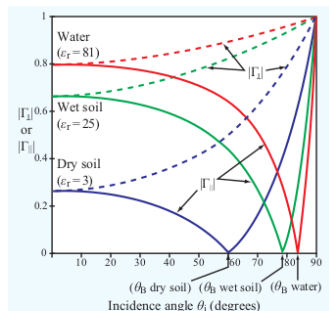
- What can be determined from just the **easier to determine parameters** θ_i , ω , μ_1 , μ_2 , ϵ_1 , and ϵ_2 ?

We could calculate the **Fresnel reflection and transmission coefficients** $\Gamma_{\perp}$, $\Gamma_{\parallel}$, $\tau_{\perp}$, $\tau_{\parallel}$, θ_r , and θ_t .

- Are these **useful** by themselves? Yes. **Why?** They tell us what fraction of the EM wave is reflected and what fraction is transmitted.

There is an angle of incidence θ_i where $\Gamma_{\parallel} = 0$ and another angle of incidence where $|\Gamma_{\perp}| = |\Gamma_{\parallel}| = 1$. The angle θ_i where $\Gamma_{\parallel} = 0$ is called the **Brewster angle**.

At **normal incidence** ($\theta_i = 0$), $\Gamma_{\perp} = \Gamma_{\parallel}$. Also, $|\Gamma_{\perp}| = |\Gamma_{\parallel}| = 1$ when $\theta_i = 90^\circ$, which is called **grazing incidence**.



Reflection and Transmission at Oblique Incidence (Cont.)

The graph shows only nonmagnetic materials. There is also a Brewster angle for $\Gamma_{\perp}$ for magnetic materials (with $\mu_1 \neq \mu_2$).

Brewster Angle

In the case of **perpendicular polarization**,

$$\Gamma_{\perp} = 0 \implies \eta_2 \cos \theta_i = \eta_1 \cos \theta_t. \quad (44)$$

Using Snell's law of refraction to replace θ_t and obtain an equation that only contains θ_i and the constitutive parameters and then solving that equation for θ_i gives the Brewster angle

$$\sin \theta_{B\perp} = \sqrt{\frac{1 - (\mu_1 \epsilon_2 / \mu_2 \epsilon_1)}{1 - (\mu_1 / \mu_2)^2}}. \quad (45)$$

What happens when $\mu_1 = \mu_2$? The denominator is 0 and $\theta_{B\perp}$ doesn't exist.

Reflection and Transmission at Oblique Incidence (Cont.)

For **parallel polarization**,

$$\Gamma_{\parallel} = 0 \implies \eta_2 \cos \theta_t = \eta_1 \cos \theta_i. \quad (46)$$

Using Snell's law of refraction to replace θ_t and obtain an equation that only contains θ_i and the constitutive parameters and then solving that equation for θ_i gives the Brewster angle

$$\sin \theta_{B\parallel} = \sqrt{\frac{1 - (\epsilon_1 \mu_2 / \epsilon_2 \mu_1)}{1 - (\epsilon_1 / \epsilon_2)^2}}. \quad (47)$$

When $\mu_1 = \mu_2$, e.g. for **nonmagnetic media**,

$$\sin \theta_{B\parallel} = \sqrt{\frac{1 - (\epsilon_1 / \epsilon_2)}{1 - (\epsilon_1 / \epsilon_2)^2}} = \sqrt{\frac{1}{1 + (\epsilon_1 / \epsilon_2)}}. \quad (48)$$

Reflection and Transmission at Oblique Incidence (Cont.)

So,

$$\theta_{B\parallel} = \sin^{-1} \sqrt{\frac{1}{1 + (\epsilon_1/\epsilon_2)}} = \tan^{-1} \sqrt{\frac{\epsilon_2}{\epsilon_1}}. \quad (49)$$

When $\theta_i = \theta_{B\parallel}$, the parallel polarized component is **completely transmitted** into Medium 2 whereas a portion of the perpendicularly polarized component is **reflected**. In this case, **all of the reflected wave is perpendicularly polarized** and the interface “polarizes” the wave. For this reason, the Brewster angle is also called the **polarizing angle**.

Reflectivity and Transmissivity

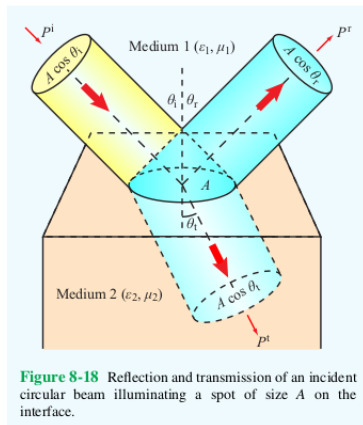
Suppose a circular beam of EM radiation shines on the boundary between two lossless media and that the area of beams at the boundary is A .

For the **perpendicularly polarized** component, the average power densities of the incident, reflected, and transmitted waves are, respectively,

$$S_{\perp}^i = |E_{\perp 0}^i|^2 / 2\eta_1, \quad (50)$$

$$S_{\perp}^r = |E_{\perp 0}^r|^2 / 2\eta_1, \quad (51)$$

$$S_{\perp}^t = |E_{\perp 0}^t|^2 / 2\eta_2. \quad (52)$$



Reflectivity and Transmissivity (Continued)

The cross-sectional areas of the beams are

$$A_i = A \cos \theta_i, \quad (53)$$

$$A_r = A \cos \theta_r, \quad (54)$$

$$A_t = A \cos \theta_t \quad (55)$$

and the average power carried by each beam is

$$P_{\perp}^i = S_{\perp}^i A_i = \frac{|E_{\perp 0}^i|^2}{2\eta_1} A \cos \theta_i, \quad (56)$$

$$P_{\perp}^r = S_{\perp}^r A_r = \frac{|E_{\perp 0}^r|^2}{2\eta_1} A \cos \theta_r \quad (57)$$

$$P_{\perp}^t = S_{\perp}^t A_t = \frac{|E_{\perp 0}^t|^2}{2\eta_2} A \cos \theta_t. \quad (58)$$

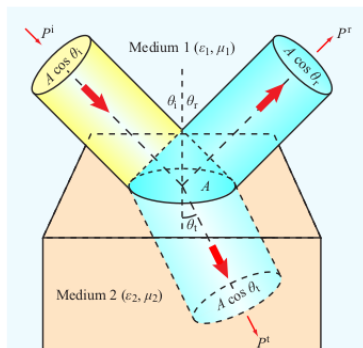


Figure 8-18 Reflection and transmission of an incident circular beam illuminating a spot of size A on the interface.

Reflectivity and Transmissivity (Continued)

The **reflectivity** (or **reflectance**) R is the ratio of the reflected power over the incident power. In this case,

$$R_{\perp} = \frac{P_{\perp}^r}{P_{\perp}^i} = \left| \frac{E_{\perp 0}^r}{E_{\perp 0}^i} \right|^2 = |\Gamma_{\perp}|^2 \quad (59)$$

since $\theta_r = \theta_i$ by Snell's laws.

In the case of **parallel polarization**, a similar argument shows

$$R_{\parallel} = \frac{P_{\parallel}^r}{P_{\parallel}^i} = \left| \frac{E_{\parallel 0}^r}{E_{\parallel 0}^i} \right|^2 = |\Gamma_{\parallel}|^2. \quad (60)$$

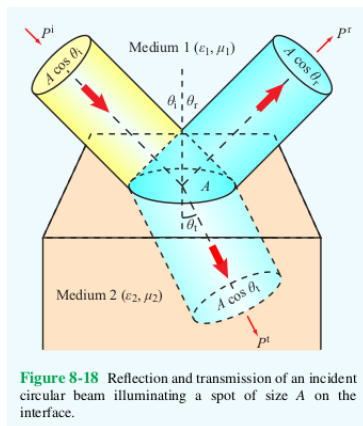


Figure 8-18 Reflection and transmission of an incident circular beam illuminating a spot of size A on the interface.

Reflectivity and Transmissivity (Continued)

The **transmissivity** (or **transmittance**) T is the ratio of the transmitted power over the incident power. In this case,

$$\begin{aligned} T_{\perp} &= \frac{P_{\perp}^t}{P_{\perp}^i} = \left| \frac{E_{\perp 0}^t}{E_{\perp 0}^i} \right|^2 \frac{\eta_1 \cos \theta_t}{\eta_2 \cos \theta_i} \\ &= |\tau_{\perp}|^2 \left(\frac{\eta_1 \cos \theta_t}{\eta_2 \cos \theta_i} \right). \end{aligned} \quad (61)$$

In the case of **parallel polarization**, a similar argument shows

$$\begin{aligned} T_{\parallel} &= \frac{P_{\parallel}^t}{P_{\parallel}^i} = \left| \frac{E_{\parallel 0}^t}{E_{\parallel 0}^i} \right|^2 \frac{\eta_1 \cos \theta_t}{\eta_2 \cos \theta_i} \\ &= |\tau_{\parallel}|^2 \left(\frac{\eta_1 \cos \theta_t}{\eta_2 \cos \theta_i} \right). \end{aligned} \quad (62)$$

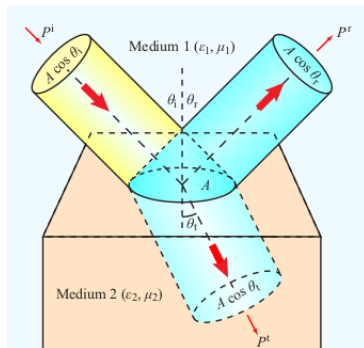


Figure 8-18 Reflection and transmission of an incident circular beam illuminating a spot of size A on the interface.

Reflectivity and Transmissivity (Continued)

Power must be conserved. So,

$$P_{\perp}^i = P_{\perp}^r + P_{\perp}^t, \quad (63)$$

$$P_{\parallel}^i = P_{\parallel}^r + P_{\parallel}^t, \quad (64)$$

which means

$$\frac{|E_{\perp 0}^i|^2}{2\eta_1} A \cos \theta_i = \frac{|E_{\perp 0}^r|^2}{2\eta_1} A \cos \theta_r + \frac{|E_{\perp 0}^t|^2}{2\eta_2} A \cos \theta_t, \quad (65)$$

$$\frac{|E_{\parallel 0}^i|^2}{2\eta_1} A \cos \theta_i = \frac{|E_{\parallel 0}^r|^2}{2\eta_1} A \cos \theta_r + \frac{|E_{\parallel 0}^t|^2}{2\eta_2} A \cos \theta_t. \quad (66)$$

Dividing these equations by their left sides gives

$$1 = \left| \frac{E_{\perp 0}^r}{E_{\perp 0}^i} \right|^2 + \left| \frac{E_{\perp 0}^t}{E_{\perp 0}^i} \right|^2 \frac{\eta_1 \cos \theta_t}{\eta_2 \cos \theta_i}, \quad 1 = \left| \frac{E_{\parallel 0}^r}{E_{\parallel 0}^i} \right|^2 + \left| \frac{E_{\parallel 0}^t}{E_{\parallel 0}^i} \right|^2 \frac{\eta_1 \cos \theta_t}{\eta_2 \cos \theta_i}. \quad (67)$$

Reflectivity and Transmissivity (Continued)

Therefore,

$$|\Gamma_{\perp}|^2 + |\tau_{\perp}|^2 \left(\frac{\eta_1 \cos \theta_t}{\eta_2 \cos \theta_i} \right) = R_{\perp} + T_{\perp} = 1, \quad (68)$$

$$|\Gamma_{\parallel}|^2 + |\tau_{\parallel}|^2 \left(\frac{\eta_1 \cos \theta_t}{\eta_2 \cos \theta_i} \right) = R_{\parallel} + T_{\parallel} = 1. \quad (69)$$

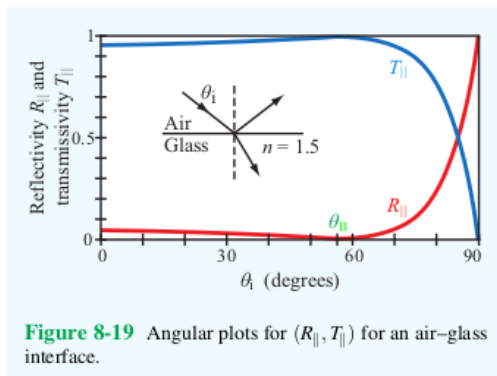


Figure 8-19 Angular plots for $(R_{\parallel}, T_{\parallel})$ for an air-glass interface.

Reflectivity and Transmissivity (Continued)

Table 8-2 Expressions for Γ , τ , R , and T for wave incidence from a medium with intrinsic impedance η_1 onto a medium with intrinsic impedance η_2 . Angles θ_i and θ_t are the angles of incidence and transmission, respectively.

Property	Normal Incidence $\theta_i = \theta_t = 0$	Perpendicular Polarization	Parallel Polarization
Reflection coefficient	$\Gamma = \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1}$	$\Gamma_{\perp} = \frac{\eta_2 \cos \theta_i - \eta_1 \cos \theta_t}{\eta_2 \cos \theta_i + \eta_1 \cos \theta_t}$	$\Gamma_{\parallel} = \frac{\eta_2 \cos \theta_t - \eta_1 \cos \theta_i}{\eta_2 \cos \theta_t + \eta_1 \cos \theta_i}$
Transmission coefficient	$\tau = \frac{2\eta_2}{\eta_2 + \eta_1}$	$\tau_{\perp} = \frac{2\eta_2 \cos \theta_i}{\eta_2 \cos \theta_i + \eta_1 \cos \theta_t}$	$\tau_{\parallel} = \frac{2\eta_2 \cos \theta_t}{\eta_2 \cos \theta_t + \eta_1 \cos \theta_i}$
Relation of Γ to τ	$\tau = 1 + \Gamma$	$\tau_{\perp} = 1 + \Gamma_{\perp}$	$\tau_{\parallel} = (1 + \Gamma_{\parallel}) \frac{\cos \theta_i}{\cos \theta_t}$
Reflectivity	$R = \Gamma ^2$	$R_{\perp} = \Gamma_{\perp} ^2$	$R_{\parallel} = \Gamma_{\parallel} ^2$
Transmissivity	$T = \tau ^2 \left(\frac{\eta_1}{\eta_2} \right)$	$T_{\perp} = \tau_{\perp} ^2 \frac{\eta_1 \cos \theta_i}{\eta_2 \cos \theta_t}$	$T_{\parallel} = \tau_{\parallel} ^2 \frac{\eta_1 \cos \theta_t}{\eta_2 \cos \theta_i}$
Relation of R to T	$T = 1 - R$	$T_{\perp} = 1 - R_{\perp}$	$T_{\parallel} = 1 - R_{\parallel}$
Notes: (1) $\sin \theta_t = \sqrt{\mu_1 \epsilon_1 / \mu_2 \epsilon_2} \sin \theta_i$; (2) $\eta_1 = \sqrt{\mu_1 / \epsilon_1}$; (3) $\eta_2 = \sqrt{\mu_2 / \epsilon_2}$; (4) for nonmagnetic media, $\eta_2 / \eta_1 = n_1 / n_2$.			

Homework

Read Chapter 8.

Homework 3 is posted and due Tuesday, May 26th,
at the beginning of class.